

The transshipment problem

The transportation problem

Michel Bierlaire

Optimization: principles and algorithms



The transportation problem

The transportation problem is somehow a transshipment problem where transshipment is not allowed.

We have suppliers and customers

There is a cost to transport items from a supplier to a customer

We need to decide how much each supplier will deliver to each customer

Electricity in Switzerland



Suppliers

- ▶ Mühleberg: 3110 GWh/year
- ▶ Beznau: 3198 GWh/year
- ▶ Leibstadt: 10205 GWh/year

Customers

- ▶ Zürich: 8961 GWh/year
- ▶ Geneva: 3777 GWh/year
- ▶ Lausanne: 2517 GWh/year
- ▶ Bern: 1258 GWh/year

Electricity in Switzerland

Unit costs

	Zürich	Geneva	Lausanne	Bern
Mühleberg	18	6	10	9
Beznau	9	16	13	7
Leibstadt	14	9	16	5



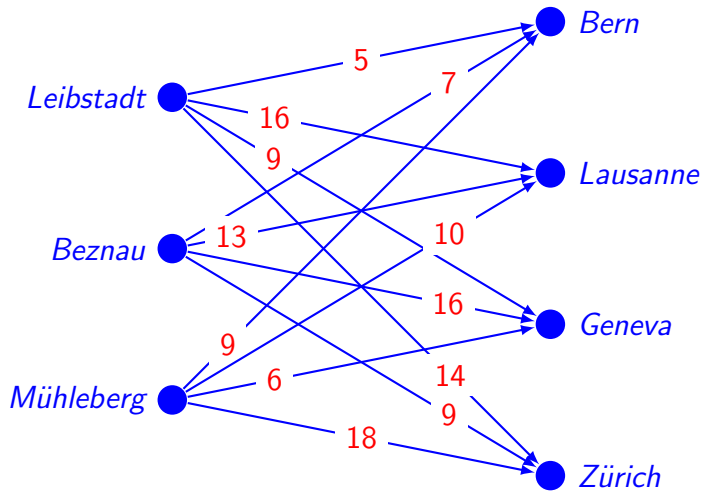
Network

Mention that the problem definition does not involve a network, contrarily to the shortest path problem and the max flow problem.

We need to create one.

Plot two or three arcs with their costs by hand.

Network



Network

Leibstadt ●

Beznau ●

Mühleberg ●

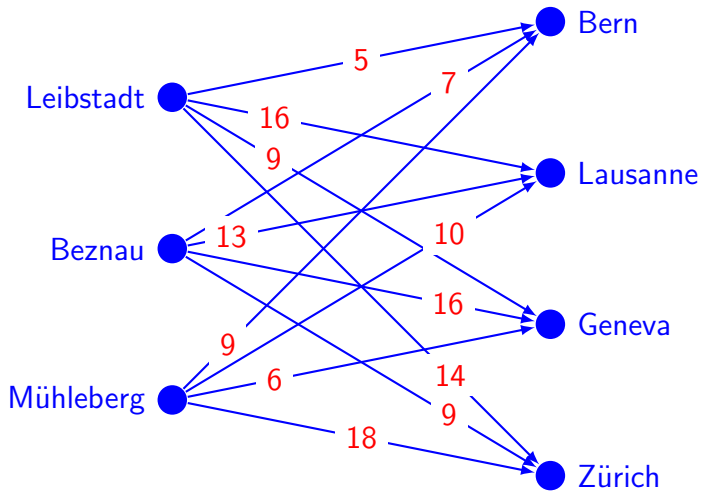
● Bern

● Lausanne

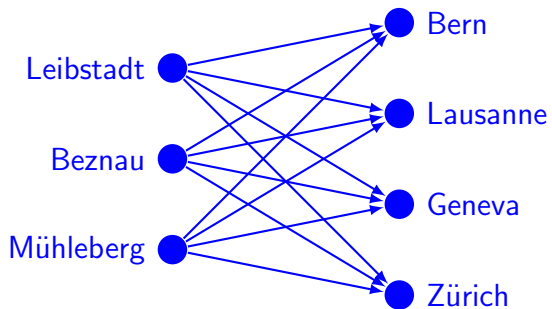
● Geneva

● Zürich

Network



Transshipment



- ▶ cost on each arc: c_{ij}
- ▶ supply for supplier nodes: s_i
- ▶ supply for customer nodes: $-t_j$
- ▶ lower bound on each arc: 0
- ▶ upper bound on each arc: $+\infty$

Transshipment

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^{m_o} \sum_{j=1}^{m_d} c_{ij} x_{ij}$$

subject to

$$\sum_{j=1}^{m_d} x_{ij} = s_i, \quad i = 1, \dots, m_o,$$

$$\sum_{i=1}^{m_o} x_{ij} = t_j, \quad j = 1, \dots, m_d,$$

$$x_{ij} \geq 0, \quad i = 1, \dots, m_o, \quad j = 1, \dots, m_d,$$

$$x_{ij} = 0, \quad \text{if } i \text{ does not serve } j.$$

Transshipment

Summary

*The transportation problem does not have an underlying network structure.
Still, it can be modeled as a transshipment problem
A conceptual network is representing the problem.
Again, it inherits the properties of the transshipment problem, such as total unimodularity.*